

Artificial Intelligence

Assignment 3

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Part 1: Game Study Selection

Title: Team-Based Game Character Ability Assignment using CSP

Description:

In a team-based strategy game, a player must assign one ability to all five characters, each having different role, before the match. Each character have role-based limitations, and certain abilities may conflict or require pairing with each other - Challenge is to assign abilities maintaining all game rules and achieving good team strength.

Goal:

Assign exactly one ability to each player such that All hard constraints (game rules) are satisfied and soft constraints (team preferences) are optimized.

Challenges (Constraints):

- Some players cannot use certain abilities
- Some abilities conflict when used together
- Some characters must have matching abilities
- Some combinations are preferred.

CSP Model:

Variables (Nodes):

C1: Tank

C2: Sniper

C3: Healer

C4: Assassin

C5: Support

Domains:

Domain for each variable:

{ Shield, Stealth, Heal, Boost }

Unary Constraints:

- Tank can't use Stealth $\Rightarrow C1 \neq \text{Stealth}$
- Healer can only use Heal or Boost
 $\Rightarrow C3 \in \{ \text{Heal}, \text{Boost} \}$

Binary Constraints:

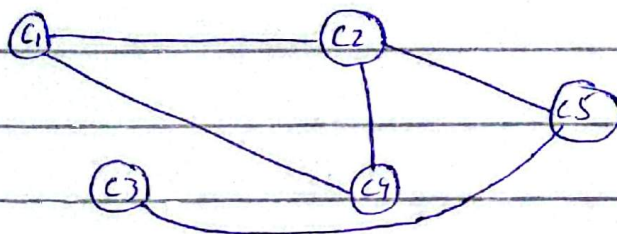
- Both Tank & Sniper can't have boost
 $\Rightarrow \text{NOT } (C1 = \text{Boost AND } C2 = \text{Boost})$

- Sniper & Assassin can't have same ability $\Rightarrow C2 \neq C4$
- Healer & Support must have same ability $\Rightarrow C3 = C5$
- Assassin & Tank cannot both use Shield
 $\Rightarrow \text{NOT } (C4 = \text{Shield AND } C1 = \text{Shield})$
- Support & Sniper must have different abilities
 $\Rightarrow C5 \neq C2$

Soft Constraint:

- Prefer at least one character to have Boost

Constraint Graph:



Part 2: Node Consistency:

C1 & C3 will have domains changed.

Before:

$C1 = \{\text{Shield, Stealth, Heal, Boost}\}$

$C2 = \{\text{Shield, Stealth, Heal, Boost}\}$

$C3 = \{\text{Shield, Stealth, Heal, Boost}\}$

$C4 = \{\text{Shield, Stealth, Heal, Boost}\}$

$C5 = \{\text{Shield, Stealth, Heal, Boost}\}$

After:

$C1 = \{\text{Shield, Heal, Boost}\}$

$C2 = \{\text{Shield, Stealth, Heal, Boost}\}$

$C3 = \{\text{Heal, Boost}\}$

$C4 = \{\text{Shield, Stealth, Heal, Boost}\}$

$C5 = \{\text{Shield, Stealth, Heal, Boost}\}$

Part 3: Arc Consistency (AC-3 Algorithm)

Initial Queue of Arcs:

$(C1 \rightarrow C2), (C2 \rightarrow C1)$

$(C2 \rightarrow C4), (C4 \rightarrow C2)$

$(C3 \rightarrow C5), (C5 \rightarrow C3)$

$(C4 \rightarrow C1), (C1 \rightarrow C4)$

$(C5 \rightarrow C2), (C2 \rightarrow C5)$

AC-3 Iterations:

Iter 1:

Arc: $(C3 \rightarrow C5)$

Constraint: $C3 = C5$

Before:

$C3 = \{\text{Heal, Boost}\}, C5 = \{\text{Shield, Health, Heal, Boost}\}$

Revise:

Heal & Boost both supported in $C5$

After:

No Change

Revision? False

Iter 2:

Arc: $(C5 \rightarrow C3)$

Constraint: $C5 = C3$

Before:

$C5 = \{\text{Shield, Stealth, Heal, Boost}\}$, $C3 = \{\text{Heal, Boost}\}$

Revise:

Remove Shield & Stealth as no match in $C3$

After:

$C5 = \{\text{Heal, Boost}\}$

Revision? True

Add Arcs:

Neighbors of $C5 \rightarrow (C2 \rightarrow C5)$

Iter 3:

Arc: $(C2 \rightarrow C5)$

Constraint: $C2 \neq C5$

Before:

$C2 = \{\text{Shield, Stealth, Heal, Boost}\}$ $C5 = \{\text{Heal, Boost}\}$

Revise:

No change needed

After:

No Change

Revision? False

Iter 4:

Arc Suggested: $(C2 \rightarrow C4)$

Constraint: $C2 \neq C4$

No change needed. (Same as above)

Iter 5:

Arg: $(C4 \rightarrow C2)$

No change

Iter 6:

Arg: $(C1 \rightarrow C2)$

Constraint: NOT $(C1 = \text{Boast AND } C2 = \text{Boast})$

No change as choice exist for $C1 = \text{Boast}$

Iter 7:

Arg: $(C2 \rightarrow C1)$

No change (same as above)

Iter 8:

Arg: $(C4 \rightarrow C1)$

Constraint: NOT $(C4 = \text{Sterile AND } C1 = \text{Sterile})$

No change as choice exist for $C4 = \text{Sterile}$

Iter 9:

Arg: $(C1 \rightarrow C4)$

No change (same as above)

Iter 10:

Arg: $(C5 \rightarrow C2)$

No change

Final Domains:

$C1 = \{ \text{Shield, Heal, Boost} \}$

$C2 = \{ \text{Shield, Stealth, Heal, Boost} \}$

$C3 = \{ \text{Heal, Boost} \}$

$C4 = \{ \text{Shield, Health, Heal, Boost} \}$

$C5 = \{ \text{Heal, Boost} \}$

→ $C5$ reduced due to $C5 = C3$ constraint

→ No empty domains → solution exist

Part 4: Backtracking Search

Step 1:

Variable Selection:

Domains:

• $C3 \& C5 = 2 \text{ values}$ • others = 3/4 values

Degree Heuristic:

$C3$ have 1 neighbor ($C5$)

$C5$ have 2 neighbor ($C3, C2$)

→ Selecting $C5$

Value Ordering (LCV for C5)

Domain = {Heal, Boost}

Heal:

Force C3=Heal, leave flexibility in C2

Boost:

Force C3=Boost, May conflict with C2 later

LCV order: {Heal, Boost}

⇒ C5 = Heal Assigned

Interference (MAC: Run AC3)

Constraint: C3 = C5

C3 = Heal

After:

C3 = {Heal}

No further pruning

Step 2:

Variable Selection:

Remaining: C3 = 1 value, others = 3/4 values

⇒ Select C3

Value Ordering (LCV for C3):

C3 = {Heal}

⇒ C3 = Heal Assigned

Interference:

C3=C5 satisfied, No change

Step 3:

Variable Selection:

Remaining: $C1 = 3$ values, $C2 = C4 = 4$ values

$\Rightarrow C1$ selected

Value Ordering (LCV for $C1$):

Domain = {Shield, Heal, Boost}

Shield:

May conflict with $C4$

Heal:

Safe, no strong restriction

Boost:

May conflict with $C2$

LCV order: {Heal, Shield, Boost}

$C1 = \text{Heal}$ selected

Inference:

No strong propagation

Step 4:

Variable Selection:

Remaining: $C2 = 4$, $C4 = 4$

$C2$ have more constraints $\Rightarrow C2$ selected

Value Ordering (LCV for $C2$):

Domain = {Shield, Stealth, Heal, Boost}

$C2 \neq C5 \Rightarrow C5 = \text{Heal}$

$\Rightarrow$ Remove Heal

Shield & Stealth $\Rightarrow$ flexible

Boast $\Rightarrow$ May effect C1

LCV order, {Shield, Stealth, Boast}

C2 = Shield selected

Inference:

Constraint: C2 $\neq$ C4

Remove Shield from C4

C4 = {Stealth, Heal, Boast}

Step 5:

Variable Selection:

C4 remains

Value ordering (LCV for C4):

Domain: {Stealth, Heal, Boast}

Stealth:

Constraint dealt above so valid

Boast:

Satisfies soft constraint

Heal:

Valid

LCV order = {Boast, Stealth, Heal}

C4 = Boast selected

Inference,

No changes needed

No backtracking required, all variables assigned.

Part 5: Final Solution.

C1 = Heal

C2 = Shield

C3 = Heal

C4 = Boost

C5 = Heal

Hard Constraints Check:

1. $C1 \neq \text{Shield}$ $\Rightarrow C1 = \text{Heal}$ ✓
2. $C3 \in \{\text{Heal}, \text{Boost}\}$ $\Rightarrow C3 = \text{Heal}$ ✓
3. $\text{NOT}(C1 = \text{Boost AND } C2 = \text{Boost}) \Rightarrow C1 = \text{Heal}, C2 = \text{Shield}$ ✓
4. $C2 \neq C4$ $\Rightarrow \text{Shield} \neq \text{Boost}$ ✓
5. $C3 = C5$ $\Rightarrow \text{Heal} = \text{Heal}$ ✓
6. $\text{NOT}(C4 = \text{Shield AND } C5 = \text{Shield}) \Rightarrow C4 = \text{Boost}$ ✓
7. $C5 \neq C2$ $\Rightarrow \text{Heal} \neq \text{Shield}$ ✓

Soft Constraint Check:

Prefer one character to have Boost $\Rightarrow C4 = \text{Boost}$ ✓